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Article in Transactions of the American Fisheries Society · May 2009

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Sablefish Predation on Juvenile Pacific Salmon in the Coastal Marine Waters of Southeast Alaska in 1999

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Abstract.—At-sea observations of predation by age-1 and older (age-1+) sablefish *Anoplopoma fimbria* on juvenile Pacific salmon *Oncorhynchus* spp. were combined with laboratory studies to determine gastric evacuation rates and were used to estimate summer predation impact in the northern region of Southeast Alaska. In June and July 1999, up to 63% of sablefish examined from trawl catches in strait habitat had each consumed one to four juvenile pink salmon *O. gorbuscha*, chum salmon *O. keta*, or sockeye salmon *O. nerka*. In two laboratory experiments, field-captured sablefish were acclimated without food in compartmentalized flow-through tanks with conditions manipulated to reflect the photoperiod and temperature regimes of summer. These sablefish were each offered one whole, preweighed juvenile chum salmon; consumption events were observed, and the sablefish were then sacrificed at predetermined time intervals. Prey biomass remaining in the stomach of each sablefish predator was weighed, and an exponential model of the decline in percent biomass over time was used to yield instantaneous evacuation rates of 0.049 at 12°C and 0.027 at 7°C. From field data combined with model-derived estimates of meal frequency, we estimated that 0.8–6.0 million juvenile salmon were consumed by age-1+ sablefish in the 500-km² area of Icy Strait during a 33-d period. Moreover, a 10-year time series of catches indicated that 1999 was a year of unusually high age-1+ sablefish abundance and relatively low juvenile salmon abundance. We speculate that sablefish predation in 1999 could have affected abundance of out-migrating juveniles and contributed to low harvests of returning adult pink salmon in 2000 and adult chum salmon in 2002. Our results suggest that sablefish predation on juvenile Pacific salmon can occur during episodic strong year-classes of sablefish and may affect adult salmon returns.

Sablefish *Anoplopoma fimbria* and Pacific salmon *Oncorhynchus* spp. are important components of marine ecosystems throughout the eastern Pacific Rim and constitute valuable socioeconomic resources for coastal communities. Sablefish are commercially fished from Baja California to the Bering Sea (Moser et al. 1994), while salmon are fished from California to the Chukchi Sea (ADFG 2007a; CDFG 2007). In Southeast Alaska, ex-vessel commercial values of these species over the past 5 years averaged US\$27 million for sablefish and \$76 million for salmon (ADFG 2007a, 2007b). However, interannual production is highly variable for both sablefish (Sigler and Fujioka 1988; King et al. 2000; Schirripa and Colbert 2006) and salmon (Coronado and Hilborn 1998; Beamish and Mahnken 2001; Wertheimer et al. 2007). Therefore, identifying mechanisms that govern the production of these fishery resources would provide critical information to fishery managers and stakeholders. Understanding life history processes and species interactions, such as predation events, may help to explain fluctuations in year-class strength and contribute to better forecasting and management of these species.

Despite their disparate natal origins, the life history transitions and movements of sablefish and salmon allow them to interact at several life history stages. Sablefish originate in the Gulf of Alaska, and the neustonic juveniles are advected onshore to epipelagic habitats in their first summer (Mason et al. 1983; Kendall and Matarese 1987; Wing and Kamikawa 1995; Rutecki and Varosi 1998a, 1998b). They remain near shore for one or more years before migrating to deeper offshore habitats (Mason et al. 1983; Dark and Wilkins 1994; Maloney and Sigler 2008) and may not become demersal until as late as age 5 (Grover and Olla 1986, 1987; Tokranov 2002). In contrast, young Pacific salmon migrate from inland freshwater habitat to marine waters in the spring (Groot and Margolis 1991). They occupy the upper 15–20 m of the water column (Jaenicke et al. 1984; Emmett et al. 2004) as they make the transition from littoral to neritic to offshore habitat in the Gulf of Alaska by fall. All species return to their natal streams within a few years via the same migratory corridors used during out-migration (Elling and Macy 1955; Nakatani et al. 1975; Orsi et al. 2004). Thus, seaward-migrating juvenile salmon are potentially vulnerable to predation by juvenile sablefish in midsummer in inside strait waters, while juvenile sablefish moving onshore are vulnerable to predation by adult salmon returning from the Gulf of

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Received July 31, 2008; accepted December 1, 2008
Published online May 7, 2009

Alaska (Wing 1985; Brodeur et al. 2004; Orsi et al. 2006). In this paper, we consider only predation by juvenile sablefish on juvenile salmon.

Predation is considered to be the largest component of mortality during early marine life of salmon that influences recruitment and survival of adult salmon (Hargreaves and LeBrasseur 1985; Moss et al. 2005; Wertheimer and Thrower 2007). The important trophic role of juvenile salmon is reflected in numerous reports of predation by a broad range of bird, marine mammal, and fish species (e.g., Hargreaves and LeBrasseur 1986; Dobrynina et al. 1989; Dolloff 1993; Scheel and Hough 1997; Nagasawa 1998; Willette et al. 1999; Orsi et al. 2007; Trites et al. 2007). The effect of this predation in freshwater systems has been studied extensively, but comparatively few estimates exist for marine piscivorous fish (Beamish et al. 1992; Emmett and Sampson 2007; Duffy and Beauchamp 2008), largely because it is difficult to obtain sufficient sample numbers on a regular basis.

Predation impact and consumption estimates also require evacuation rate data (e.g., Elliott and Persson 1978; Olson and Mullen 1986) in conjunction with complementary field data. Evacuation rate data are often lacking for sablefish and other species that are difficult to maintain in the laboratory (Sullivan and Smith 1982; Rindorf 2004). As a result, evacuation rates from similar predators feeding on different prey may be substituted from literature values (e.g., Livingston 1983; Albers and Anderson 1985; Cartwright et al. 1998); additionally, these predators may come from different environments. Temperature is thought to be the most critical factor regulating evacuation rates (Persson 1986; Bromley 1994; Jobling 1994; Héroux and Magnan 1996) and also affects estimates of predation impact. Therefore, better information on evacuation rates, trophic links, and predation processes could improve our understanding of marine ecosystems.

During routine annual monitoring of coastal epipelagic habitats by the Southeast Coastal Monitoring (SECM) project of the Alaska Fisheries Science Center's Auke Bay Laboratory in 1999, age-1 and older (age-1+) sablefish co-occurred in high abundance with juvenile salmon in the northern region of Southeast Alaska. Shipboard stomach analyses revealed that many of the sablefish ate juvenile salmon (Orsi et al. 2000a, 2000b). Because sablefish catches and predation on juvenile salmon have been rare events since the SECM began in 1997 (Murphy et al. 1999; Orsi et al. 1998, 2000b, 2007), we wanted to explore the potential importance of this interaction. We speculated that an episode of abundant juvenile sablefish could reduce juvenile salmon abundance

and affect the number of returning adults. The purpose of our study was to combine at-sea observations of predation with a laboratory evacuation study to estimate the effect of age-1+ sablefish predation on juvenile Pacific salmon during their first summer at sea. Objectives for the field component of our study were to determine the rate of sablefish predation on juvenile salmon and the number, species composition, and size selection of salmon in the sablefish diet. The objective for the laboratory component of our study was to estimate evacuation rates for age-1+ sablefish fed single juvenile salmon at two temperatures. We then used our long-term data set of annual catches to graphically explore the relationship of sablefish abundance with concurrent juvenile salmon abundance and subsequent adult salmon harvest.

Methods

At-sea sampling and processing.—Sablefish were collected with a Nordic 264 rope trawl using the National Oceanic and Atmospheric Administration (NOAA) ship *John N. Cobb* at nine stations in the vicinity of Icy Strait and Chatham Strait in the northern region of Southeast Alaska during June–September 1999 (Figure 1; Orsi et al. 2000a). These collections were made during the annual SECM studies of early marine ecology and potential predators of juvenile salmon in inshore, strait, and coastal habitats (Orsi et al. 2001, 2007; Park et al. 2004; LaCroix et al., in press). Fish were captured during daylight hours over several days near the end of each month while fishing the trawl (~24-m spread) with the headrope at the surface and the footrope at approximately 20-m depth. Water temperature profile data were collected with a Seabird SBE 19 conductivity–temperature–depth profiler to describe the thermal range of the fish.

Fish captured in each trawl haul were anesthetized, sorted, identified, and counted. Juvenile salmon were routinely measured with an electronic measuring board (nearest mm fork length [FL]) and then were labeled and frozen individually; large catches were subsampled. Frozen individuals were later weighed in the laboratory (nearest g wet weight [WW]) to estimate prey biomass and for use in evacuation experiments. Sablefish and other potential predators were similarly measured and weighed on board the vessel. A subsample of stomachs (from ≤ 20 sablefish/haul) was excised, and the total content weights (W) were estimated by subtracting the empty stomach weight from the full stomach weight. Volumetric percent composition (%V) of fish and invertebrate prey was visually estimated and then converted to percent weight (%W = $W \times \%V$). Individual prey categories were not weighed due to processing time constraints. Fish prey

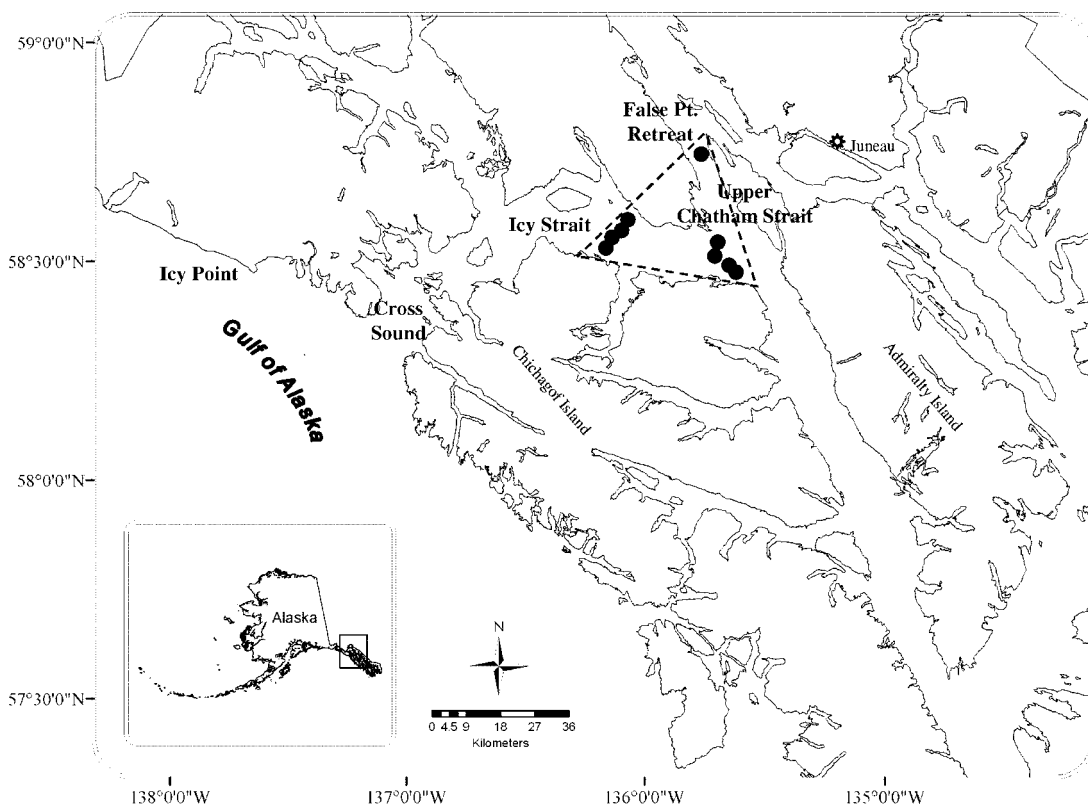


FIGURE 1.—Sampling localities for the Southeast Coastal Monitoring program in the northern region of Southeast Alaska, June–September 1999. Sablefish predation on juvenile salmon was observed in the 500-km² area in the vicinity of Chatham and Icy straits (bounded by triangle). Trawling was conducted from the National Oceanic and Atmospheric Administration ship *John N. Cobb*. Inset shows location of study area in Southeast Alaska.

were identified to species when possible, and intact specimens were measured for total length or FL.

Data summary and analysis.—Catches of sablefish and juvenile Pacific salmon were summarized by total numbers and catch per unit effort (CPUE; average number of fish per trawl haul) for the strait predation area by month. Sablefish prey composition was summarized by month as %W and percent frequency of occurrence of juvenile salmon, other identifiable fishes, unidentifiable fish remains, and invertebrates. Mean FL, WW, and relative sizes of salmon and sablefish (percent body length [% BL] and percent body weight [% BW]) were summarized by month in the catch and for predators. Sablefish selection for salmon species was tested by comparing frequencies in the stomachs with frequencies in the catch (chi-square test for independence, test statistic = χ^2 ; Minitab 2003). Prey size selection was tested by comparing the species' lengths in stomachs with their lengths in the catch (*t*-test). For pink salmon *O. gorbuscha* and chum salmon *O. keta*, we also checked for differences in size

distributions in the stomachs versus the catch using plots of cumulative frequency of occurrence (increments of 5 mm FL) and tested for differences in mean size between the species (*t*-test). Sizes of field and experimental sablefish were compared using analysis of variance (ANOVA).

Evacuation experiment trials.—To examine the evacuation rates of sablefish, we captured live fish in the vicinity of Juneau, Alaska, by rod and reel (Rutecki and Meyers 1992) or surface rope trawl (Orsi et al. 2000b) between July and October 1999. Viable sablefish with minimal hook- or net-induced injuries were transported in tanks supplied with fresh, flowing seawater to the Auke Bay Laboratory by the *John N. Cobb* or the RV *Quest*. Captive fish were held outdoors in covered circular tanks (715-L capacity; $n \leq 21$ fish). Fish were fed chopped herring and squid for up to 2 months before the experiments and then were fasted for at least 3 d before the experimental trials began.

Experiments were conducted at 12°C ($n = 30$ sablefish) and 7°C ($n = 24$ sablefish) in indoor,

covered, fiberglass rectangular tanks (503-L capacity) during September and November 1999. The tanks were sectioned into three compartments using screens; circulating pumps with air stones were installed on each end. Experimental temperatures represented summer field conditions at the surface and at 20-m depth (Orsi et al. 2000b, 2004; Park et al. 2004). Seawater was delivered by the laboratory's intake pipe at 25-m depth in Auke Bay and only required heating for the September experiment (ambient temperature $\approx 9^\circ\text{C}$). Photoperiods for both experiments were adjusted to mimic midsummer conditions: daylight from 0430 to 2130 hours, 1 h of dusk before and after daylight, and 6 h of darkness.

Individual sablefish were randomly assigned to a tank compartment with a preassigned digestion time treatment (1, 2, 4, 6, 8, 16, 32, 48, 72, or 96 h) and then were allowed to acclimate for approximately 24 h. In each experimental trial, each sablefish was fed one previously frozen, whole juvenile chum salmon; this meal standard was chosen for simplicity and because predator samples were limited. At least three replicate samples were obtained per digestion interval. Thawed prey from the previous summer rope trawl surveys were measured and weighed and then were introduced at the compartment surface. Each sablefish was allowed 1 h to feed and was replaced if it did not consume the prey ($\sim 10\%$ of the time). The moment of prey consumption was recorded as the start time. After ingesting prey, sablefish were left undisturbed for the digestion interval before being killed, measured, and weighed. Postmortem stomach sampling was done by methods similar to those onboard the vessel. For each sablefish, we computed initial prey % BL and % BW as well as the percent of prey weight remaining (%W) after the digestion interval.

To model the evacuation process over time and estimate an instantaneous rate of prey evacuation, the replicate %W values for each temperature–time interval were plotted against time and an exponential curve (e.g., Elliott and Persson 1978; Persson 1986; Dwyer et al. 1987) was fit to the data as follows:

$$W_t = W_0 e^{-rt},$$

where W_0 is the initial weight of food, W_t is the percent weight of food remaining after t hours, and the slope r represents the instantaneous evacuation rate (per h). We tested model fits of r for separate temperatures versus a single rate for both temperatures using the likelihood ratio test, and we estimated the 95% confidence intervals (CIs) of the slope parameters from their likelihood profiles (Hilborn and Mangel 1997).

Estimation of sablefish predation impact on juvenile

salmon.—Summer predation impact was estimated for an area of approximately 500 km² in the northern region of Southeast Alaska using SECM data from all nine stations along Icy Strait and Chatham Strait (Figure 1). The temperature-specific predation impact formula was:

$$\text{Predation impact}_T = (D \times 500 \text{ km}^2) \times \text{PR} \times \text{NC} \\ \times \text{MF} \times [(33 \text{ d}) \times 24 \text{ h/d}],$$

where T = temperature, D = mean sablefish density, PR = predation rate, NC = mean number consumed by an individual, and MF = sablefish meal frequency (h). We calculated D using CPUE and the sample area per trawl. Trawl area was estimated by multiplying the horizontal spread (24 m) by the distance fished (1 nautical mile = 1.852 km) during a standard 20-min period at 3 knots ($0.024 \text{ km} \times 1.852 \text{ km} = 0.044 \text{ km}^2/\text{haul}$). Predator density D (number/km²) was computed by dividing the June–July mean CPUE by this trawl area, and the total predator population was then estimated using a multiplier of 500 for the total strait area. Using only the fish with food in their stomachs (Temming and Mergardt 2002; Rindorf and Lewy 2004), we calculated PR as the percent frequency of salmon-eating sablefish at two levels. The minimum PR was calculated from the frequency of direct observations of salmon in sablefish stomachs. The maximum PR was calculated from the frequency of salmon and unidentified fish remains that were potentially salmon; we used the ratio of salmon eaters among sablefish with identifiable remains to estimate the additional number of salmon-eating sablefish from those that contained only unidentifiable remains. Three levels of MF were estimated from the evacuation curves by extrapolating the time (h) required to evacuate 50, 75, and 90% of the original prey biomass at each temperature; these different MFs were used to account for hypothetical differences in individual motivation to feed, which could affect the overall population consumption. We assumed a 33-d predation season based on the interval between our first and last observations at sea (June 26 to July 29, 1999), and we calculated the number of meals possible based on MF. We estimated a range of predation impact and then calculated the percentage effect of D when it varied by 1 SE of the mean and the percentage effect of NC when it varied between the low and high number of salmon encountered in sablefish stomachs.

We used the 10-year time-series of SECM catch data to speculate on the implications of predation impact in 1999. We graphically compared the annual abundance of age-1+ sablefish with juvenile and adult salmon abundances using (1) the June–July CPUE of juvenile

TABLE 1.—Catches of age-1 and older (age-1+) sablefish and juvenile salmon in 23 trawl hauls conducted within a 500-km² area near Icy Strait, Alaska, June–August 1999. Data include the number of fish caught, percent of total juvenile salmon catch contributed by each salmon species (% number), mean fork length (FL, mm; measured fresh), mean wet weight (WW, g; fresh sablefish, thawed salmon), and relative size of single salmon expressed as a percentage of sablefish predator body length (BL) or body weight (BW). In July, sizes of all species were estimated from subsamples of the large total catches shown.

Month and variable	Age-1+ sablefish	Pink salmon	Chum salmon	Sockeye salmon
Jun				
Number	5	32	477	94
% number	—	5	79	16
Mean FL (SE)	309 (8)	97 (3)	103 (1)	126 (2)
Mean WW (SE)	309 (23)	7 (1)	11 (1)	21 (2)
Prey % BL	—	31.4	33.3	41
Prey % BW	—	2.4	3.4	6.6
Jul				
Number	146	588	683	35
% number	—	45	52	3
Mean FL (SE)	340 (1)	113 (1)	126 (1)	133 (4)
Mean WW (SE)	444 (5)	16 (1)	21 (1)	26 (2)
Prey % BL	—	33	36.5	39
Prey % BW	—	4	4.6	6
Aug				
Number	5	14	14	2
% number	—	75	30	2
Mean FL (SE)	372 (3)	139 (4)	130 (4)	122 (23)
Mean WW (SE)	541 (30)	26 (5)	20 (7)	20 (9)
Prey % BL	—	38	36	39
Prey % BW	—	5	4	6

pink salmon and chum salmon (the most abundant species and the most common in sablefish stomachs) and the CPUE of juvenile sablefish from our study area and (2) the annual commercial harvests of adult pink salmon and chum salmon for the Southeast Alaska region. We assumed that the high abundance of age-1+ sablefish in straits during 1999 indicated an episode of widespread abundance of the 1998 year-class in Southeast Alaska (Sigler et al. 2001; Hanselman et al. 2007) and that the sample area represented the region. Pink salmon invariably return after 1 year at sea, and the majority of chum salmon return at ocean age 3 (Groot and Margolis 1991). We obtained salmon harvest data from the Alaska Department of Fish and Game (ADFG 2007a) for Southeast Alaska (excluding Yakutat, which was considered outside the study area) and lagged these data by 1 year (for pink salmon) and 3 years (for chum salmon) after the year of juvenile out-migration. Harvest data are presented only through 2004 because final values for ocean age-3 chum salmon harvests in Southeast Alaska for subsequent years were not yet available.

Results

Juvenile Sablefish and Juvenile Salmon Interaction

In total, 86 trawl hauls were completed during SECM surveys from June to September 1999 in inshore, strait, and coastal habitats (Figure 1; Orsi et al. 2000b). However, age-1+ sablefish and juvenile Pacific salmon co-occurred only in strait habitat in

June, July, and August; they did not co-occur in coastal habitat, and sablefish were virtually absent from inshore habitat (Orsi et al. 2000b). In June, more than 600 juvenile pink salmon, chum salmon, and sockeye salmon *O. nerka* were caught with five age-1+ sablefish, while in July (when CPUE was highest) more than 1,300 juvenile salmon were caught with 146 sablefish (Table 1). In August and September, catches of juvenile salmon declined as their distribution shifted offshore into the Gulf of Alaska; they were uncommon in strait habitat when the last few sablefish were caught in August. Salmon and sablefish distributions did not completely overlap during the June–July predation period. Juvenile salmon were caught with sablefish in 33% of the 18 hauls and without sablefish in 44% of the hauls. Sablefish were caught without salmon in one haul, but one individual sablefish had eaten three pink salmon. Neither sablefish nor salmon were caught in three of the hauls. Juvenile pink salmon and chum salmon were much more abundant than juvenile sockeye salmon, which also peaked earlier, in June (Table 1). The July sample from False Point Retreat had the highest catches of pink salmon, chum salmon, and sablefish.

Sablefish were strongly piscivorous throughout the summer, but juvenile salmon were consumed only in June and July (Table 2). Salmon constituted more than 80% of prey weight in a diet that totaled 92–98% fish in both months. Salmon were consumed by more than half of the sablefish, more frequently than any other

TABLE 2.—Observations of salmon-eating, age-1 and older (age-1+) sablefish from trawl catches collected within the 500-km² area near Icy Strait, Alaska, during June–July 1999 and descriptions of juvenile salmon prey found in their stomachs. Data include the number of fish observed, percent total number of juvenile salmon eaten (% number), mean fork length (FL, mm), mean wet weight (WW, g), and relative size of salmon prey expressed as a percentage of sablefish predator body length (BL). Salmon prey were not individually weighed. No sablefish preyed on juvenile salmon in August (see Table 1).

Month and variable	Age-1+ sablefish	Prey contained in stomachs		
		Pink salmon	Chum salmon	Sockeye salmon
Jun				
Number	2 ^a	0	2	1
% number	—	0	67.6	33.3
Mean FL (SE)	315 (12)	—	98 (1)	145
Mean WW (SE)	340 (40)	—	—	—
Mean prey % BL (SE)	—	—	31 (1)	45
Jul				
Number	22 ^b	15	25	2
% number	—	35.7	59.5	4.8
Mean FL (SE)	346 (2)	114 (4)	118 (2)	166 (22)
Mean WW (SE)	454 (14)	—	—	—
Mean prey % BL (SE)	—	33 (1)	34 (1)	47 (6)

^a Four of the five sablefish examined in June contained prey; two of these ate salmon.
^b Forty-four of the 51 sablefish examined in July contained prey; 22 of these ate salmon.

prey species, and at all nine stations (Figure 1). Predation was most common in July. In June, four of the five sablefish examined contained food: two ate salmon alone or with other fish, one ate only other species, and one contained only unidentifiable fish remains. In July, 44 of the 51 sablefish examined contained food: 22 ate salmon alone or with other species, one contained only other species, eight contained only unidentifiable fish remains, and 14 contained only invertebrates. In August, three of the five sablefish examined contained food: none ate salmon, one contained only unidentifiable fish remains, and two contained only invertebrates. Other fishes identified in the stomachs included sculpins (*Cottidae*), Pacific herring *Clupea pallasii*, capelins *Mallotus villosus*, wolf-eels *Anarrhichthys ocellatus*, walleye pollock *Theragra chalcogramma*, and fish larvae.

A range of high and low PRs was calculated for the June–July sablefish samples. The minimum PR estimated was 50% (24 of the 48 feeding sablefish). The maximum PR estimated was 63% (30 of 48 sablefish, including six of the nine sablefish that contained only unidentifiable fish remains; these were categorized as salmon eaters based on the 24:39 ratio of salmon eaters observed among all sablefish with identifiable fish prey). Overall, the mean NC per sablefish was 1.8 juvenile salmon (1.5 in June and 1.9 in July), but higher or lower NC values were frequently observed (nine sablefish had one salmon, eight had two salmon, six had three salmon, and one had four salmon).

All salmon in sablefish stomachs were identified to species, and no juvenile Chinook salmon *O. tshawytscha* or coho salmon *O. kisutch* were found. Sablefish

consumed chum salmon most frequently (60% of the 45 salmon specimens identified); pink salmon and sockeye salmon constituted 33% and 7% of salmon specimens, respectively. Chum salmon and sockeye salmon were consumed in both June and July, but pink salmon were consumed only in July, when their abundance was high; no salmon were consumed in August, when abundance was low for all species (Tables 1, 2). In July, when juvenile salmon were most abundant and most predation events were observed, sablefish selected salmon species in proportion to their abundance ($\chi^2 = 1.835$, $df = 2$, $P = 0.399$); therefore, chum salmon were eaten most commonly. However, sablefish were size selective for chum salmon that were smaller than the mean size available in the trawl catches (consumed size [mean $\pm$ SE] = 118 $\pm$ 2 mm; size in trawl catch = 126 $\pm$ 1 mm; t -test: $P = 0.003$); the size frequency of chum salmon prey was also shifted to the left of the catch size frequency, confirming selection for smaller fish than were available. In contrast, pink salmon prey size did not differ from their size in trawls (consumed size = 114 $\pm$ 4 mm; size in trawl catch = 113 $\pm$ 1 mm; t -test: $P = 0.389$), and pink salmon were significantly smaller than chum salmon in the catch ($P = 0.000$). The few juvenile sockeye salmon consumed were larger than the average size in the trawls (Tables 1, 2).

The sizes of both sablefish and juvenile salmon increased over the summer, but relative prey size showed no trend. Mean size of sablefish increased approximately 20% by length and 75% by weight, while mean size of juvenile pink salmon and chum salmon increased approximately 40% by length and

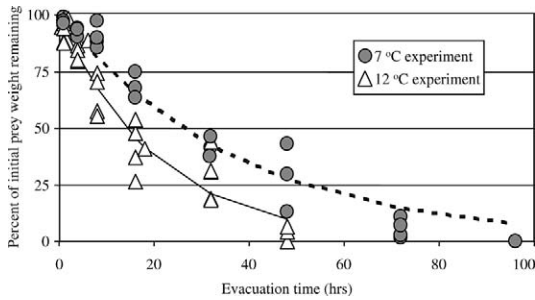


FIGURE 2.—Percentage of consumed prey biomass remaining over time in the stomachs of age-1 and older (age-1+) sablefish ($n = 54$) that were each fed a single juvenile chum salmon in experiments conducted at 7°C and 12°C. The decline in prey biomass over time is described by the exponential model; instantaneous evacuation rate was 0.027 (95% confidence interval [CI] = 0.026–0.028) at 7°C and 0.049 (95% CI = 0.048–0.050) at 12°C.

greater than 50% by weight (Table 1). Sockeye salmon were larger and more variable in size than pink salmon and chum salmon. Sablefish stomach capacity was suggested by the relative prey sizes (Table 2). Maximum length capacity for a single prey could be inferred from the sizes of large, uncommon sockeye salmon, which averaged 45–47% BL and up to approximately 7% BW. A more common prey size was observed for pink salmon and chum salmon, which averaged 31–34% BL and about 2–5% BW (Tables 1, 2). Therefore, a meal of four juvenile salmon could represent up to approximately 20% BW. This potential ration was corroborated by stomach contents of salmon-eating sablefish caught at sea; prey items averaged around 7% BW and as much as 19% BW.

Laboratory Predation Experiments and Evacuation Rates

Temperature clearly affected sablefish r -values (Figure 2). Observations on the decline of initial prey

% BW for 7°C rarely overlapped those for 12°C. Digestion was significantly slower at the colder temperature (likelihood ratio test: $P < 0.001$); r was 0.027 (95% CI = 0.026–0.028) at 7°C compared with 0.049 (95% CI = 0.048–0.050) at 12°C. Prey species was readily discernible for 4–8 h after ingestion at 12°C and for up to 16 h at 7°C. Sablefish used in the 7°C experiment were larger than those used in the 12°C experiment (Table 3), and both groups were larger than sablefish examined at sea a few months earlier (Table 1; ANOVA, Tukey's pairwise comparisons: $P = 0.000$). However, the single-fish meal sizes in both laboratory experiments were comparable with those of field-sampled sablefish, averaging about 4% BW (Table 3). We estimated that sablefish would feed from about every 14 h to 3.5 d depending on r and how empty the stomachs were before feeding motivation returned. The times estimated for 50, 75, and 90% evacuation of prey were 26, 51, and 85 h, respectively, at 7°C and 14, 29, and 47 h, respectively, at 12°C (Figure 2, Table 4). Thus, MF would be greater for sablefish needing less time to digest prey in the warmer environment, particularly if feeding resumed when stomachs were only half empty.

Sablefish Predation Impact on Juvenile Salmon

Predation impact in 1999 was estimated only from data collected during the months and locations where there was evidence of sablefish predation on salmon (i.e., June and July in strait habitat; Tables 1, 2). Estimated CPUEs (mean $\pm$ SE) were 8.4 ± 6.5 age-1+ sablefish/haul and 106.1 ± 37.6 juvenile salmon/haul (pink salmon, chum salmon, and sockeye salmon combined). Thus, the average sablefish population (95,328 fish) was roughly one-tenth of the juvenile salmon population (1,205,177 fish) in the 500-km² area. These populations could vary by approximately 77% and 35%, respectively, based on their SE values. Likewise, the total number of salmon consumed could vary by 40% if NC was allowed to vary from the

TABLE 3.—Numbers, mean fork length (FL, mm), and mean wet weight (WW, g) of age-1 and older sablefish predators and juvenile chum salmon prey, and relative size of prey expressed as a percentage of predator body length (BL) or body weight (BW), during experiments conducted at two temperatures in 1999. Lengths and weights were measured on fresh predators and thawed prey; each predator was given a single prey item.

Variable	7°C		12°C	
	Sablefish	Chum salmon	Sablefish	Chum salmon
Number	25	25	30	30
Mean FL (SE)	392 (3)	123 (2)	367 (5)	122 (1)
Mean WW (SE)	504 (14)	19 (1)	439 (17)	18 (1)
Mean prey % BL (SE)	—	31.3 (0.5)	—	33.4 (0.6)
Mean prey % BW (SE)	—	3.7 (0.2)	—	4.3 (0.3)

TABLE 4.—Potential impact (number consumed) of predation by age-1 and older (age-1+) sablefish on juvenile salmon within a 500-km² area encompassing Icy Strait, Alaska. Estimates are based on a fixed, 33-d period of sablefish–salmon interaction in June–July 1999, a mean of 1.8 salmon/stomach, and a population estimate of 95,328 sablefish; variables include low (7°C) and high (12°C) temperatures that determine evacuation rate, three meal frequency (MF) intervals (h between meals) determined by return of feeding motivation (from experimental times to 50, 75, and 90% evacuation of original prey biomass at 7°C or 12°C), and low versus high rates of predation (% of sablefish that consumed salmon). Estimates of predation impact could further vary by about 40–77% if either the number of prey per predator or the sablefish population differed from the averages (see text).

Percent of original meal (prey biomass) evacuated	MF (h)	Number of meals	Total number eaten per predator	Total number consumed under scenario of:	
				Low-impact, 50% predation	High-impact, 63% predation
7°C temperature					
50	26	30.5	54.5	2,598,942	3,248,678
75	51	15.5	27.8	1,324,951	1,656,189
90	85	9.3	16.7	794,971	993,713
12°C temperature					
50	14	56.6	101.3	4,826,607	6,033,259
75	29	27.3	48.9	2,330,086	2,912,608
90	47	16.9	30.2	1,437,713	1,797,141

average of 1.8 juvenile salmon/sablefish to the frequently observed value of 1 or 3 salmon/sablefish. Based on the average populations and average number of prey per gut, predation impact varied 7.5-fold (Table 4). The low estimate in the 7°C environment was derived from a range of approximately 9–31 meals/predator and 17–55 prey consumed/predator in the time period examined (33 d). The high estimate in the 12°C environment was derived from a range of 17–57 meals/predator and 30–101 prey consumed/predator. Combined with the two PRs, these variables yielded potential consumption impacts of approximately 0.8–6.0 million juvenile salmon eaten (Table 4). At an average of 16.8 g per individual juvenile salmon over the time period, this represents an average biomass of approximately 13–101 metric tons of salmon consumed.

Plots of age-1+ sablefish CPUE and juvenile pink salmon and chum salmon CPUEs in the strait area showed that in 1999, sablefish had the highest abundance observed in the 10-year SECM time series (Figure 3a) and juvenile salmon abundance was relatively low (Figure 3b, c). This was most pronounced in July, typically the month of greatest juvenile salmon abundance (Orsi et al. 2000b, 2004, 2007). Sablefish were absent in most years. Commercial catches of adult pink salmon in 2000 and adult chum salmon in 2002 (Figure 3b, c) were also low compared with most years. We speculate that the 1999 episode of high sablefish abundance and predation could have contributed to the relatively low abundance of juvenile salmon in that year and to the relatively low commercial harvests of returning salmon that were observed later.

Discussion

Juvenile Sablefish and Juvenile Salmon Interaction

We report new evidence of age-1+ sablefish predation on juvenile Pacific salmon, the first published study of *r* in sablefish, and an estimate of the potential impact of a marine piscivore on juvenile salmon. While the sablefish–salmon interaction is uncommon, the high incidence of sablefish in the Icy Strait area in 1999 was unusual and the PR was the highest we have observed for any species in our 10 years of stomach sampling from surface trawl catches. This predation could exert important effects on salmon because sablefish coincided with the peak abundance of salmon in straits during their seaward migration to the Gulf of Alaska. Both juvenile salmon abundance and adult harvest vary widely among years. Significant predation is difficult to isolate from field observations (e.g., Beamish and Neville 2001) or to link to variation in abundance because of the numerous environmental factors that accumulate to affect survival. Our results suggest that sablefish predation on juvenile salmon can occur during episodic strong year-classes of sablefish and may affect adult returns. We speculate that sablefish predation in 1999 could have affected abundance of out-migrating juveniles in the northern region of Southeast Alaska and contributed to relatively low harvests of returning adult pink salmon in 2000 and adult chum salmon in 2002.

Interactions of sablefish and salmon in Southeast Alaska can vary interannually because both exhibit strong shifts in abundance. Sablefish recruitment is reported to be highly variable and to exhibit periodic large year-classes (e.g., Sasaki 1985; Sigler and Fujioka 1988; King et al. 2000) where juveniles are

widespread (McFarlane and Beamish 1992; Rutecki and Varosi 1998a; M.F.S., unpublished data). Salmon abundance has also increased dramatically since enhancement facilities were developed in the 1970s, coupled with recent high ocean survival of both wild and hatchery stocks (Downton and Miller 1998; Beamish and Mahnken 2001). Harvests of chum salmon and pink salmon were lower before and higher after the 1976–1977 shift in climate regime (Wertheimer 1997), whereas strong sablefish year-classes in 1977–1981 were followed by a period of relatively low recruitment (Hanselman et al. 2007). This suggests that abundances of sablefish and salmon are not necessarily correlated.

In our study, age-1+ sablefish were most abundant and were predominantly piscivorous in coastal waters of the northern portion of Southeast Alaska during summer 1999. More than half of the sablefish sampled consumed juvenile salmon, which dominated their diets in the Icy Strait area in June and July. Sablefish had fed recently and often had 2–3 salmon/stomach and up to 4 salmon/stomach, in contrast to many piscivores that capture few prey daily (Breck 1993). Salmon size ratio was favorable to these sablefish (Juanes 1994; Mittelbach and Persson 1998; Scharf et al. 1998); the high PR, selection for the most abundant species, and size selectivity suggest high encounter rates and capture efficiencies (Hargreaves and LeBrasseur 1986; Werner et al. 1996; Galarowicz and Wahl 2005). Although salmon are unusual fish prey, sablefish are piscivorous from age 0 (Conway 1967; Cailliet et al. 1988; Grover and Olla 1990; Sigler et al. 2001). Sablefish consume many fish species throughout their extensive range (e.g., Shubnikov 1963; Grinols and Gill 1968; McFarlane and Beamish 1983; Orlov 1997; Tanasichuk 1997; Yang and Nelson 2000), and their diet occasionally includes juvenile or adult salmon (Orlov 1997; Tanasichuk 1997; Orsi et al. 2000a; Hanselman et al. 2007). Anecdotes of juvenile sablefish predation on juvenile salmon have also been reported in Prince William Sound (pink salmon fry: L. Haldorson, University of Alaska, personal communication) and Southeast Alaska (Chinook salmon smolts: R. Vallion, Medvije Hatchery, Northern Southeast Regional Aquaculture Association, personal communication).

Sablefish did prey on other fish species besides juvenile salmon in our study. However, we do not know whether low abundances of other species led to increased predation on juvenile salmon because our daytime trawl hauls do not sample these vertically migrating fishes (Orsi et al. 2004, 2006, 2007). Most sablefish caught in our study had fed within several hours and during daylight, when salmon are the

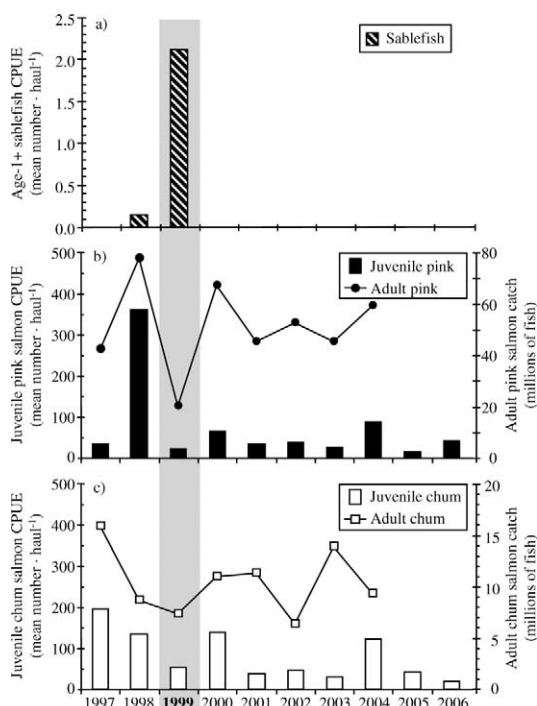


FIGURE 3.—Abundances in 1999 (gray-shaded section) compared with interannual trends from 1997 to 2006 in Southeast Alaska for (a) age-1 and older (age-1+) sablefish, (b) juvenile and adult pink salmon, and (c) juvenile and adult chum salmon. Sablefish and juvenile salmon abundances are expressed as June–July catch per unit effort (CPUE; mean number of fish per haul) in strait habitat of the northern region of Southeast Alaska; adult salmon data are commercial harvests (catch; millions of fish) from all of Southeast Alaska except the Yakutat fishery. Adult harvests are lagged by 1 year for pink salmon and 3 years for chum salmon to coincide with the out-migration year in which juveniles could have interacted with age-1+ sablefish.

dominant forage species in surface waters up to 20 m depth (Orsi et al. 2004, 2007). The incidence of forage species in sablefish stomachs in 1999 did not indicate that sablefish used this much more abundant resource at night (Orsi et al. 2004), when their visual threshold is low (Ryer et al. 2002; De Robertis et al. 2003). Even if none of the unidentifiable fish remains in sablefish stomachs were salmon, salmon were eaten more than twice as often overall as other species.

Sablefish consume high percentages of fish by volume, biomass, or frequency (e.g., Shubnikov 1963; Sasaki 1985; Tanasichuk 1997; Yang and Nelson 2000), and their abundance has occasionally been linked to other fish species (Laidig et al. 1997; Orlov 1997). Nonetheless, we found only two estimates of sablefish consumption impact on fish stocks in the literature. Along the west coast of Canada,

juvenile sablefish annually consumed 35,000 metric tons of fish, mostly Pacific herring (McFarlane and Beamish 1983), while adult sablefish consumed an estimated 1,000–7,000 metric tons (Ware and McFarlane 1986). Our estimate of 13–101 metric tons of predation impact on juvenile salmon is comparatively low because the sablefish were younger, smaller predators and because the juvenile salmon were small prey compared with adult Pacific herring. In addition, the predation period was shorter, and both the predation area and predator population were smaller. However, our estimated effect on juvenile salmon numbers was high (0.8–6.0 million fish consumed). Ware and McFarlane (1986) concluded that juvenile sablefish had less long-term effect on British Columbia Pacific herring stocks than did other piscivores, although sablefish were important predators during a strong year-class.

Evacuation Rate of Sablefish in Comparison with Other Marine Species

Our study found that sablefish r -values (0.027 at 7°C and 0.049 at 12°C) were generally comparable with values published for other piscivorous, coldwater demersal species. Sablefish r -values increased with temperature, similar to those of gadoids, an important coldwater fish group (Tyler 1970; Dwyer et al. 1987; Smith et al. 1989; Knutsen and Salvanes 1999; Rindorf 2004). Sablefish were also like gadoids in having long gut residence times for meals (e.g., Jones 1974; Smith et al. 1989; Paul et al. 1990; Pillar and Barange 1995; Temming and Mergardt 2002). Sablefish r -values were more similar to the r -values of fish-eating walleye pollock, Pacific hake *Merluccius productus*, Pacific cod *Gadus macrocephalus*, and spiny dogfish *Squalus acanthias* than to the r -values of crustacean-eating individuals of those species at similar temperatures in the northeast Pacific Ocean (Dwyer et al. 1987; Smith et al. 1989; Paul et al. 1990; Tanasichuk et al. 1991; Beamish et al. 1992). Gastric emptying times on the order of days are not unusual for coldwater demersal species and may be reported in lieu of r . Comparable with Atlantic cod *G. morhua*, sablefish required 2.0–3.5 d to empty their stomachs at similar temperatures (Karpevich and Bokova 1937; Tyler 1970; Daan 1973). A few gadoids took only 12–48 h to empty stomachs (Jones 1974; Bromley 1988).

In addition to temperature, predation impact of sablefish in our study was strongly affected by MF. The three MF values we used bracketed the point where the evacuation curves flattened and suggested a stimulus to feed (~75% evacuated). Other studies have shown that gadoids are more likely to ingest a new meal as biomass of the previous meal decreases (e.g.,

Magnusson and Aspelund 1997; Rindorf 2002; Temming and Mergardt 2002). It was apparent that individual sablefish fed before stomachs were completely empty because successive meals of salmon were obvious in some stomachs. Stomach fullness also varied from empty to distended among sablefish individuals. Piscivores that feed infrequently tend to have such variable feeding rhythms (Elliott and Persson 1978; Knutsen and Salvanes 1999; Pedersen 2000; Temming and Mergardt 2002), and assuming various intervals between sablefish meals allowed us to construct a range of population predation impacts.

Although sablefish r -values and meal residence times were similar to those of gadoids, their predation impact could be greater because of their large stomach capacity (Tyler 1970; Paul et al. 1990; Bromley 1991). Stomach capacity can indirectly affect r because it limits meal volume and the number and size of prey consumed (Breck 1993; Knutsen and Salvanes 1999; Temming and Mergardt 2002) and is related to feeding rhythms (Ruggerone and Rogers 1984; Turesson et al. 2006). Further studies on sablefish evacuation patterns would be useful to determine how factors such as stomach capacity, return of appetite, meal size, and prey size, number, and density interact to influence predation impact.

It is possible that our experimental r -values did not accurately reflect those of the field-caught sablefish because of size differences between field and laboratory predators (Jones 1974; Smith et al. 1989). However, several authors reported no effect of fish size on r (Tyler 1970; Elliott 1991; Andersen 1999). The size differences of sablefish in our study are to be expected because of growth during the 2–5 months that elapsed between field and laboratory experiments, but all sablefish were in their second year of life. The use of larger predators in the experiments was logistically unavoidable, and we have no information on size effects because our experiments were only designed to examine temperature effects.

Evacuation rates of the field-caught sablefish could also differ from those of the experimental fish because most of the field-caught fish contained more than one juvenile salmon. Thus, the total % BW of experimental meals was lower than that of meals for most field-caught sablefish, even though the relative size of a single prey was similar. Previous research concerning the effect of meal size on r and feeding rhythms has not shown consistent results. Piscivores may evacuate smaller meals at slower (Andersen 2001) or faster (Tyler 1970; Paul et al. 1990) rates than, or at the same rate as, larger meals (Bromley 1988; Knutsen et al. 1999). The effect of single- and multiple-prey meals has also been debated (Andersen and Beyer 2007), but

many studies have shown that multiple prey eaten successively will evacuate at an overall rate similar to that of a single-prey meal (Ruggerone 1989; Bromley 1991; Elliott 1991; Magnusson and Aspelund 1997; Temming and Mergardt 2002). For these reasons and the fact that nearly 38% of the sablefish from field collections did contain only one salmon, the values of r from our laboratory experiments can reasonably be applied to estimate field predation.

Sablefish Predation Impact on Juvenile Salmon

Our analysis using variable temperatures, MFs, and predation frequencies, along with the average predator population size and average number of salmon eaten, indicated that the impact of age-1+ sablefish predation on juvenile salmon could vary 7.5-fold. Meal frequency had the greatest single effect; when MF was increased by requiring less stomach emptying before feeding motivation returned (50% instead of 90%), predation impact approximately tripled at either study temperature. Temperature had the next-greatest effect. The 5°C temperature increase from 7°C to 12°C during evacuation approximately doubled predation impact. Increasing the predation incidence from 50% to 63% of the population produced a 20% increase in predation impact. By comparison, varying either the predator population or NC could reduce or increase the estimate by up to 77% and 40%, respectively.

For 1999, we estimated that 1.2 million juvenile salmon were present on average during June–July and that an estimated 0.8–6.0 million salmon were consumed by sablefish predators. This predation impact is feasible because hundreds of millions of salmon migrate through Icy Strait for several months during each year (Orsi et al. 2001, 2004), and our abundance estimates do not account for emigration, immigration, or other mortalities from the larger, initial population size. Our estimate is also feasible because it is derived from local field observations combined with laboratory r -values that are consistent with those of other cold-water fishes. The summer CPUE, peak month, and the most abundant salmon species in the strait area vary interannually, but sablefish predation patterns in 1999 reflected the species composition, relative abundance, and seasonal timing of juvenile salmon in trawl catches (Orsi et al. 2000b; J.A.O., unpublished data); chum salmon were preyed on earlier and more frequently than pink salmon, sockeye salmon were rarely eaten, and coho salmon and Chinook salmon were never eaten. These observations reflect differences in predation vulnerability reported for juvenile salmon species and sizes (Healey 1982; Hargreaves and LeBrasseur 1985, 1986; Magnhagen 1988; Abrahams and Healey 1993; Moss et al. 2005). Previous studies have shown

that common influences in the early life histories of pink salmon, chum salmon, and coho salmon can lead to positive correlations between adult catches (Downton and Miller 1998; LaCroix et al., in press). Our catch data do not support sablefish predation as a factor contributing to other years of either low juvenile salmon abundance or low adult harvests.

Predation incidence and impact on juvenile salmon are likely to be linked to the abundance and relative timing of predator and prey arrivals (Thorsteinson 1962; Willette et al. 1999; Svenning et al. 2005). The Icy Strait corridor is important for marine mammals and fish that migrate seasonally (e.g., Nakatani et al. 1975; Orsi et al. 2000a; Matkin et al. 2007; Trites et al. 2007), suggesting that small fish passing through the strait could “run the predator gauntlet.” Several studies have shown that fish and other predators are attracted to high densities of juvenile salmon during their out-migration, such as at river mouths or hatchery release sites (e.g., Ruggerone and Rogers 1984; Beamish et al. 1992; Scheel and Hough 1997). Although our study was conducted when wild fish had moved off the beach and survived the most vulnerable nearshore phase and when hatchery fish were distant in both time and space from their release sites, juvenile salmon may remain vulnerable because the many stocks are “funneled” through Icy Strait from the broad spatial distribution of Southeast Alaska inside passages. By contrast, predation was not observed in inshore and coastal habitats, where fish were more dispersed and where both the catches and salmon–sablefish overlap were lower.

This prominent intersection of the 1998 year-class of sablefish with juvenile salmon was an unusual, well-timed opportunity for predation. Rather than an annual event, it may have resulted from environmental conditions that favored survival or that altered onshore movement timing so as to place sablefish in strait habitat with salmon. Sablefish recruitment has been linked to oceanographic conditions that affect food and transport. Warm El Niño years were associated with inferior prey composition and larval recruitment compared with cool La Niña years (Grover and Olla 1987; Schirripa and Colbert 2006). Years of warm sea surface temperatures, strong southwesterly winds, and intense Aleutian lows were associated with stronger recruitment of juveniles (McFarlane and Beamish 1992; King et al. 2000). Our observations took place in the middle of the 1998–2000 La Niña cycle that occurred between El Niño cycles (NOAA Climate Data Center 2008), but annual surface temperatures in inside waters of Southeast Alaska were cooler in 1999 than in 1998 (Park et al. 2004; J.A.O., unpublished data). The delayed effects of eddies and other water mass movement induced by El Niño–Southern Oscillation

events (Sigler et al. 2001) could have affected sablefish distribution in inside straits. Sablefish are also known to modify their behavior according to food availability (Kennedy 1974; Sullivan and Smith 1982). However, catches of either age-0 or age-1+ sablefish have been rare in our trawls regardless of annual temperature patterns, sablefish year-class strength, or salmon abundance (Park et al. 2004; Orsi et al. 2000a, 2007; Hanselman et al. 2007) and do not suggest that sablefish movements are regularly timed to juvenile salmon migrations in the Icy Strait area. In years that sablefish and salmon do intersect, environmental conditions such as temperature could also affect predation impact through salmon migration timing (Orsi et al. 2000a, 2001) and growth rates (Mortensen et al. 2000; Wertheimer et al. 2004; Seo et al. 2006) as well as sablefish feeding rates (Sogard and Olla 1998, 2001; Stoner and Sturm 2004).

Our estimate of sablefish predation impact could be affected by their depth distribution and behavior. Sablefish feeding necessarily took place in the 20-m layer of surface waters of the straits because juvenile salmon reside there. However, because of their variable vertical migration behavior (Orlov 1997; Rutecki and Varosi 1998a, 1998b; Sogard and Olla 1998, 2001), juvenile sablefish may not remain at the surface between meals and may experience other conditions depending on their behavior after predation events. Any salmon digested by sablefish below 20-m depth would not be detected or accounted for in our estimates of predation impact. In addition, the r -values we described are pertinent if sablefish do not descend to colder waters, where r would decrease and assimilation would increase. We note that sablefish have not been captured in the occasional years when SECM also conducted night sampling (Orsi et al. 2004, 2007).

The depth distribution and vertical movement patterns of age-1+ sablefish in the straits are unknown, and we recognize that our trawl surveys may not represent sablefish abundance. The SECM trawl is widely used for juvenile salmon studies along the Pacific coast (Orsi et al. 2007), but other species may not be sampled representatively because of their depth distribution or size. The unusual occurrence of sablefish in surface waters of Icy Strait in 1999 remains unexplained, and the sablefish population could be considerably higher than we estimated. Nonetheless, our comparisons are useful because they are based on long-term observations and sampling with the identical trawl and vessel at consistent times and locations each year (Orsi et al. 2007). Our findings therefore highlight the importance of long-term studies for capturing infrequent, episodic ecological events.

Acknowledgments

We express our gratitude to A. Moles for time and assistance with laboratory setup and to our "hook-and-liners," T. Rutecki, C. Lunsford, D. Courtney, B. Wing, D. Mortensen, K. Munk, and the command and crew of the NOAA ship *John N. Cobb*, for fish collections. We thank E. Fergusson for creating the map figure and S. Heintz (Alaska Department of Fish and Game, Ketchikan) for providing additional salmon harvest data and information. Special thanks to our enthusiastic junior laboratory assistants: N. Johnson, S. Johnson, N. Orsi, K. Sigler, and A. Sigler. Reference to trade names does not imply endorsement by the U.S. Government.

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